Quality Management

10 — Light and illumination: making the defect visible

Prof. Gianluca Rossi • **dr. Lorenzo Capponi**
BSc in Engineering Management • *Università degli Studi di Perugia*
A.Y. 2026/2027

Today

The camera can only measure what the light shows

An image depends on three things: the **light**, the lens and the sensor. The last two lectures covered the lens and the sensor. Today the first — the one that decides whether a defect is in the image at all.

1. **Light meets the part**
   reflection, contrast, how much light
2. **Light sources**
   LED, halogen, fluorescent, laser, fibre guides
3. **Geometry**
   bright and dark field, coaxial, dome, backlight
4. **Special techniques**
   colour, polarisation, UV, infrared, strobes
5. **Choosing the light**
   from the defect to the setup

Why the light comes first

No algorithm can find a defect the light has not shown

The goal of lighting is **contrast** between what you look for and everything else:

$$C = (I_{max} - I_{min}) \,/\, (I_{max} + I_{min})$$

From 0 (nothing to see) to 1 (black on white). Good lighting makes the defect bright on dark or dark on bright, and keeps everything else uniform. It is chosen **before** the camera and the software — and it is the cheapest part to get right.

> **Before choosing, know**
>
> monochrome or colour image · the area and shape to light · the part's colour, finish and roughness · how fast it moves · which detail matters · how long the light must last.

What bad lighting does

- **Reflections and hot spots** that saturate the sensor and hide what is under them
- **Shadows** that look like edges
- **Low contrast**: the defect is there, at almost the same grey as its surroundings
- **Blooming** next to bright spots
- **Low signal-to-noise**: too little light, too much gain
- **Drift**: the light changes over the day, and every threshold with it

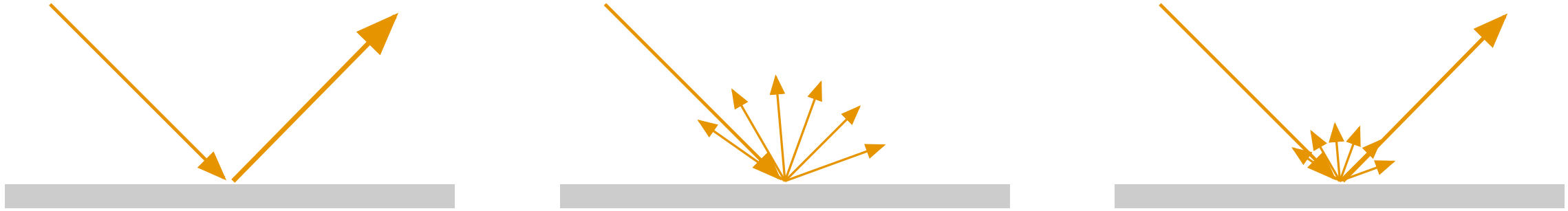
PART 1

Light meets the part

Reflection · Transmission · How much light

How a surface returns light

The finish of the part decides where the light goes



Specular — polished metal, glass
Angle out = angle in. The camera sees the light only if it sits on that one direction: **geometry decides everything**.

Diffuse — paper, matt plastic
Light scattered in every direction. The part looks the same from anywhere: easy to light, little relief.

Mixed — most real parts
A diffuse part plus a specular peak. A scratch changes the local angle and sends light elsewhere: that is what makes it visible.

Light can also be **transmitted** (glass, liquids, thin plastic) or **absorbed**. A part's colour is what it does not absorb: a red part returns red light and absorbs the rest — the basis of choosing the colour of the light (slide 9).

How much light reaches the part

Distance and angle, before any lens

For a small source of intensity I, at distance r and angle θ from the surface's normal, the illuminance is

$$E = I \cdot \cos\theta \,/\, r^2$$

- **Inverse square:** twice the distance, a quarter of the light.
- **Cosine:** light arriving at a grazing angle is spread over a larger area — at 60° from the normal, half the light.
- A large panel or a dome is not a small source: its light falls off far more slowly with distance, which is one reason it lights more evenly.

MOVING A SPOT LIGHT

A spot moved from 30 to 60 cm gives the part a quarter of the light: the exposure must grow four times, or the gain — with its noise.

UNIFORMITY ACROSS THE FIELD

A lamp 30 cm above the centre of a 300 mm field is 34 cm from its edge, at 27°: the edge gets **0.72** of the centre's light. A fixed threshold then finds edges in different places across the field.

PART 2

Light sources

Halogen · Fluorescent · LED · Laser · Fibre-optic guides · Optics that shape the light

Comparing light sources

Spectrum, stability, speed, lifetime, heat, cost

	Halogen	Fluorescent	LED	Laser
How it makes light	a tungsten wire heated white-hot; the halogen gas keeps the bulb clean	mercury vapour emits UV; a coating turns it into visible light	electrons and holes recombining in a semiconductor junction	stimulated emission: one wavelength, in phase
Spectrum	broad, visible to infrared, close to daylight	visible, in peaks	narrow (20–40 nm) per colour; white from blue + phosphor	a single line
Stability	good on DC	**flickers at 100 Hz** on the mains: bands in the image	**very stable** on DC	stable
Flash / strobe	no	no	**yes, down to microseconds**	yes
Lifetime	a few thousand hours	about 10 000 h	**about 50 000 h**	long
Heat, size	hot; small source, ideal for fibre guides	large tubes	small; needs a heat sink	small
In machine vision	fibre illuminators, legacy	rarely	**the standard**	lines, dots, patterns: 3D

Why LEDs won: bright, cheap, long-lived, stable, fast enough to strobe, in every colour from UV to infrared, and easy to build into any shape — rings, bars, domes, panels.

LEDs, and choosing the colour of the light

A monochrome camera plus the right colour of light can separate colours

- **Colour from the material:** the semiconductor's band gap sets the wavelength — red, green, blue, UV and infrared LEDs exist. White LEDs are blue LEDs with a phosphor. Red came first (1962); the efficient blue LED (1993) made white LEDs possible — Nobel Prize 2014.
- **Output:** the optical power grows with the current: $P = \eta \cdot (I/q) \cdot h\nu$. So LEDs are driven by a **constant-current** supply, not a voltage.
- **Heat** lowers the output and shifts the colour: LED bars sit on aluminium heat sinks, and a light used for measurement is warmed up before calibrating.
- **Strobe mode:** short pulses at several times the rated current give a bright flash that freezes motion.

The colour rule

A part looks **bright** under light of its own colour and **dark** under the complementary colour. Red print on a white box: under red light it vanishes, under green light it turns black.

To see…	light it with…
a red mark, dark	green or blue
a red mark, gone	red
fine detail	short wavelength (blue): less diffraction
under colour prints	near infrared

Match the light to the sensor too: silicon is most sensitive in the red and near infrared.

Lasers

Not for lighting a scene: for drawing lines, dots and patterns on it

LASER = light amplification by stimulated emission of radiation. Three properties, three uses:

Property	Use in quality control
One wavelength	a narrow filter on the lens blocks all other light
A narrow beam	lines and dots for laser triangulation — the most common industrial 3D sensor (the 3D shape measurement lecture)
Coherence: waves in phase	interferometry, for surface form and roughness down to nanometres

A **cylindrical** (or Powell) lens spreads the beam into a line. Diode lasers are used for inspection; high-power gas and solid-state lasers cut and weld.

Speckle

Coherent light on a rough surface gives a grainy pattern of bright and dark spots. It is noise for imaging — one reason lasers are not used to light a scene.

Safety: laser classes (IEC 60825-1)

Class 1: safe in normal use · Class 2: visible, protected by the blink reflex · Class 3R and 3B: hazardous to the eye · Class 4: also skin and fire. Above class 2, a production line needs enclosures, interlocks and trained staff.

Fibre-optic light guides

The source in one place, the light where it is needed

A fibre guides light by **total internal reflection**: a glass core with a slightly lower-index cladding around it. Light enters only inside a cone set by the **numerical aperture**:

$$\mathrm{NA} = \sqrt{(n_{\mathrm{core}}{}^2 - n_{\mathrm{cladding}}{}^2)} = \sin\theta_{\max}$$

- **Why use them:** the hot source (halogen or high-power LED) stays away from the part; the light comes out as a ring, a line or a spot; nothing electrical near the product.
- **Limits:** light is lost at every joint and along the length — beyond about 5 m they are inefficient and the cables expensive.

Fibre	Core	Used for
Multimode, step index	large; many paths	light guides and bundles
Multimode, graded index	index falls from the centre; paths equalised	short data links
Single mode	a few µm; one path only ($V = 2\pi a \cdot \mathrm{NA}/\lambda < 2.405$)	telecom, fibre sensors

Connectors: SMA on illuminators and spectrometers; FC, ST and SC in data links; **FC/APC**, cut at 8°, where back-reflections must be avoided — in fibre sensors.

Coherent fibre bundles and GRIN (graded-index) rod lenses also carry *images*: they are the optics of borescopes, used to inspect inside engines, bores and pipes.

Optics that shape the light

The building blocks inside every illuminator

Element	What it does	Where you find it
Diffuser	scatters light into all directions	in front of every LED panel, dome and backlight
Fresnel lens	a thin lens made of concentric steps	spotlights, and collimating large backlights
Cylindrical lens	focuses in one direction only	line lights, laser lines, barcode scanners
Beam splitter	half-mirror: lets half through, reflects half	**coaxial lighting** (slide 17)
Mirrors and prisms	redirect the view or the light	seeing the sides and back of a part with one camera
Polariser	passes one direction of polarisation	removing glare (slide 22)
Wave plate	changes the type of polarisation, e.g. linear to circular	polarisation setups, optical isolators
Filter	passes a band of wavelengths	on the lens, to see only your own light

Recap TAKEAWAYS

Parts 1 and 2 — light, surfaces, sources

1. **Lighting creates contrast:** $C = (I_{max} - I_{min})/(I_{max} + I_{min})$. It is chosen before camera and software.
2. **Surfaces:** specular, diffuse or mixed. On shiny parts the geometry of light and camera decides what is seen.
3. **How much light:** $E = I \cos\theta / r^2$. Twice the distance, a quarter of the light; large sources light more evenly.
4. **LEDs are the standard:** stable, long-lived, strobable, any colour. Choose the colour of the light to make the defect dark or bright.
5. **Lasers draw lines and patterns** for 3D, not scenes; fibre guides move the light; diffusers, Fresnel and cylindrical lenses, beam splitters and polarisers shape it.

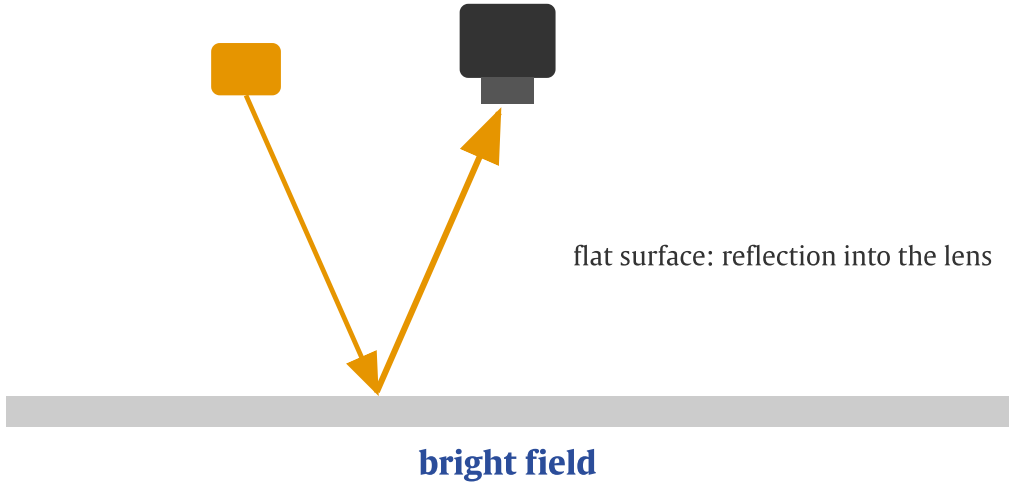
PART 3

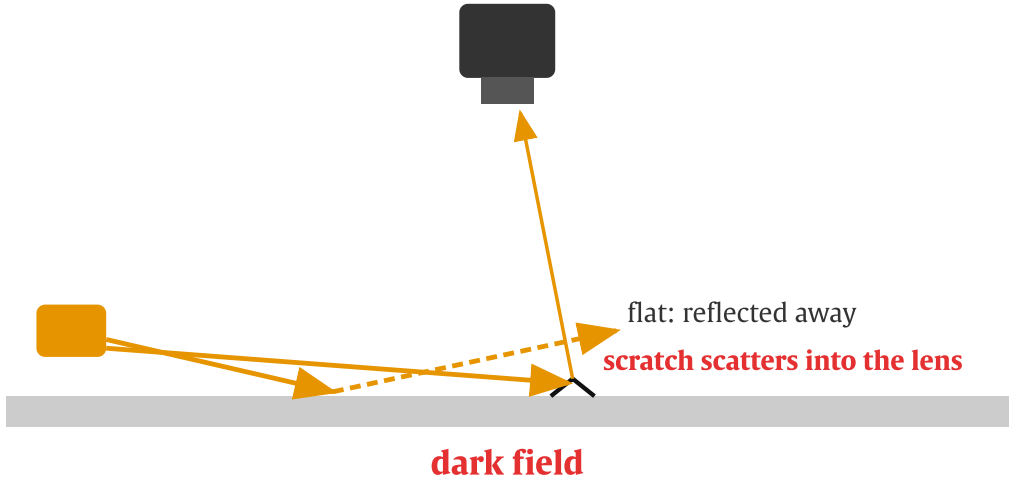
Lighting geometry

Bright and dark field · Ring · Coaxial · Dome · Backlight · Line lights

Bright field and dark field

Does the mirror reflection go into the camera, or past it?





Bright field
The reflection of a flat surface enters the camera: the surface is **bright**, and anything that tilts the light away — a dent, an engraving — is dark. Good for matt parts and flat shiny ones.

Dark field
Light at a grazing angle: a flat surface reflects it away and looks **dark**. Scratches, dust, cracks and embossed characters scatter part of it into the camera and shine **bright** on black.

Directional light and ring lights

The most common illuminators, and their angles

Directional: spot, bar, area
Intense and simple. Shadows show relief and deep features; on shiny parts it makes hot spots. A small source far away gives a more uniform field.

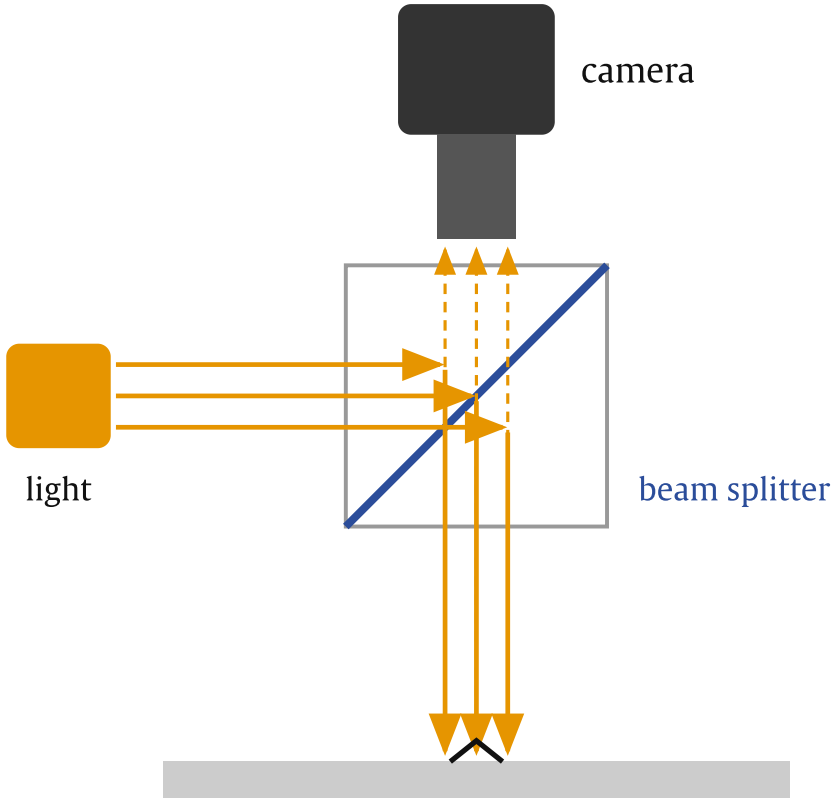
Ring light, around the lens
Light from all around the axis: almost no shadows, low contrast, suits most matt parts — the general-purpose choice. **Segmented** rings switch sectors on and off to bring out relief in one direction.

Ring angle	Gives	Shows
near the axis	bright field	print, colour, matt surfaces
angled, 30–60°	bright field with some relief	general inspection
low angle, almost grazing	**dark field** all round	scratches, engravings, edges of shiny parts

Rings with several concentric rows combine the three: one illuminator, three lightings, selected in software.

Coaxial light

Light along the camera's own axis, through a beam splitter



flat: back into the lens · engraving: dark

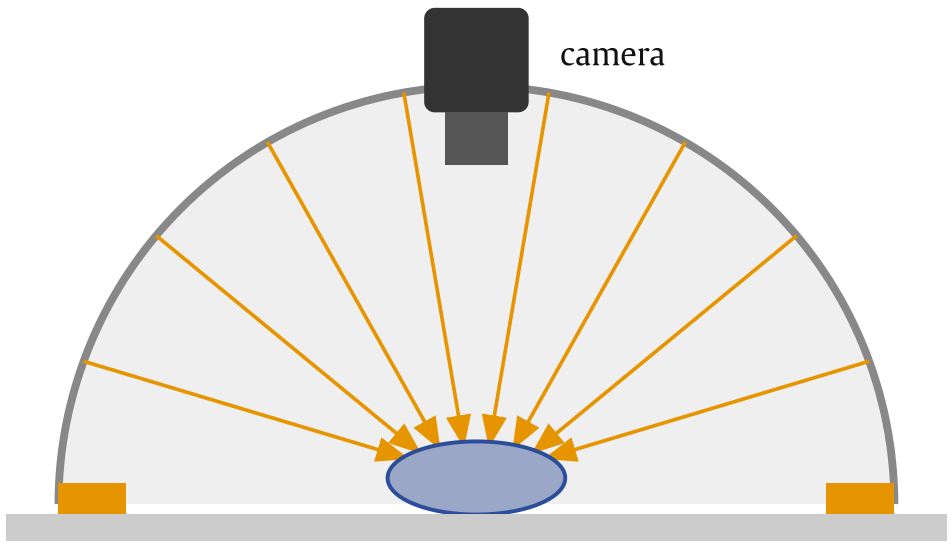
- A source at the side, a **beam splitter** at 45° in front of the lens: the light reaches the part along the axis, and the part's mirror reflection goes straight back into the camera.
- Flat shiny areas are **bright**; anything tilted — an engraving, a scratch, a shallow dent — reflects elsewhere and is **dark**.
- **Directional** coaxial (parallel rays) gives the strongest contrast; **diffuse** coaxial (a diffuser in front of the source) gives an even image without shadows.
- The beam splitter costs light: at most a quarter of the source's light comes back to the camera.

Used for

Engraved and laser-marked metal · printed circuit boards · labels and codes on shiny surfaces · polished wafers and glass.

Diffuse light: domes and tunnels

A cloudy day, in a box



light from the whole dome: no shadows, no hot spot

- A white **dome** lit from below its rim sends light onto the part from every direction, as the sky does on an overcast day: no shadows and no sharp reflections, even on curved shiny parts.
- The camera looks through a hole at the top, which leaves a dark spot in the reflection. The **cloudy-day illuminator** fills it with coaxial light, so the part is lit from every angle.
- **Tunnel** (or tube) lights do the same for long parts, in one direction.

THE HARDEST PART TO LIGHT: A BALL BEARING

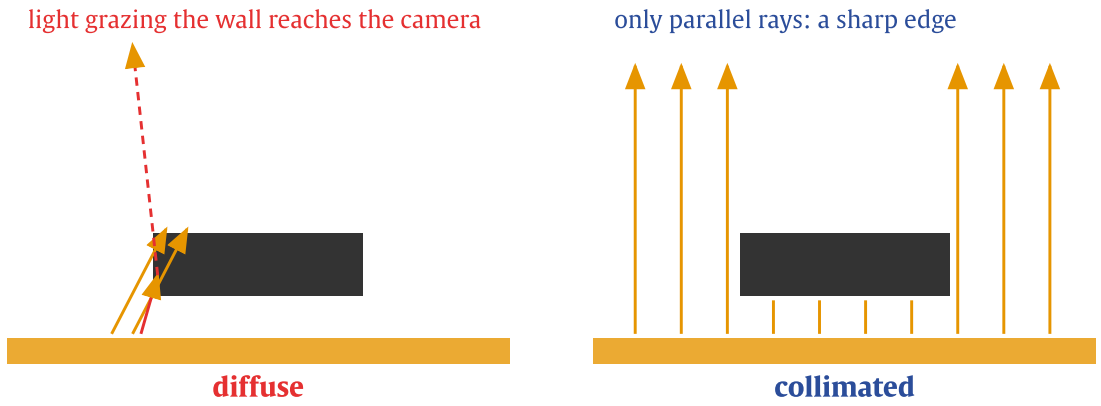
Under a ring or coaxial light, a polished ball mirrors the illuminator. Only a cloudy-day dome lights the whole sphere without showing a reflection of the lamp.

The price: diffuse light also hides relief and fine texture — exactly what dark field shows.

Backlight

The part between the light and the camera: a silhouette

- The part is black on white: the highest contrast there is, and independent of its colour and finish. The choice for **outlines, dimensions and holes** — and for transparent objects: liquid level in bottles, bubbles, inclusions in glass.
- **Area** backlights for area cameras, **line** backlights for line-scan cameras.
- A **diffuse** backlight sends light at all angles: on a thick or rounded part, some of it grazes the side wall and reaches the camera. The edge looks blurred, and the part looks smaller than it is.
- A **collimated** backlight sends only parallel rays. With a telecentric lens (lecture 7) the edge stays sharp and in place: the setup for precision gauging. A short wavelength — red or blue LEDs — keeps diffraction at the edge small.



THE WASHERS IN THE LAB

A phone screen under a sheet of paper is a diffuse backlight. On a 1.6 mm washer, part of the edge error you will see comes from it — calibrating on a master washer cancels most of it.

The same part under four lightings

An engraved metal plate — which light shows the engraving?

Lighting	Flat metal	Engraving	Verdict
Ring, near the axis	uneven, with hot spots	barely visible	poor
Coaxial	uniformly bright	**dark**, sharp	best for reading it
Low angle (dark field)	dark	**bright**, with every scratch	best for its depth and for scratches
Angled, one side	shaded	visible with shadows	shows relief, but unevenly

The lesson

There is no best light, only the best light **for one feature**. A part with several features to inspect often needs several lightings, switched in sequence, one image each.

Try before you buy

Illuminator makers publish image galleries and online lighting advisors, and lend demo kits: test on real parts, including the worst ones.

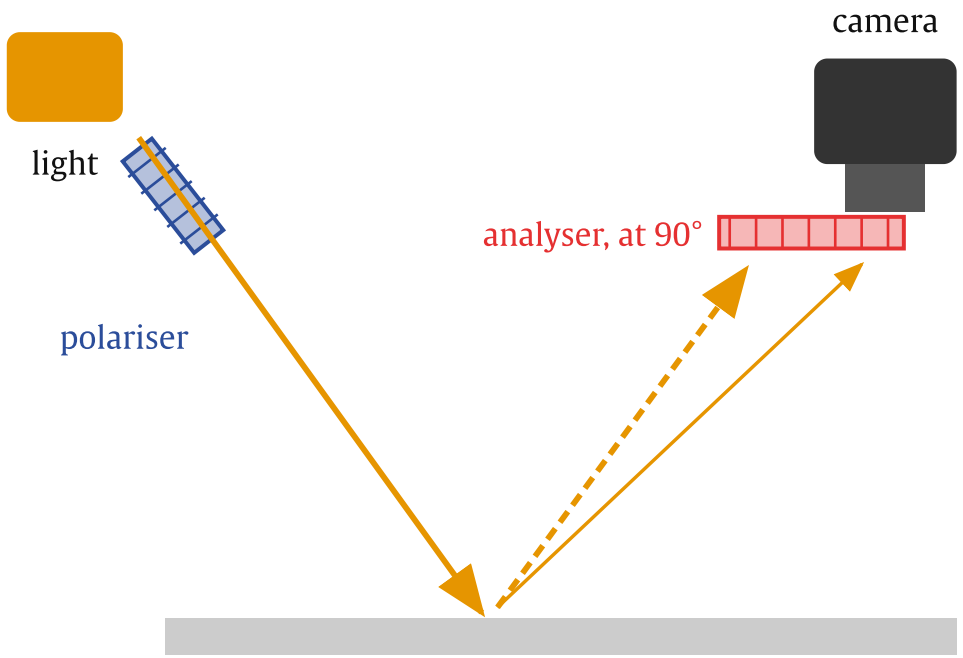
PART 4

Special techniques

Polarisation · Ultraviolet and fluorescence · Infrared · Structured light · Strobes

Polarised light

Removing glare without moving the light



mirror reflection: polarisation kept → blocked
diffuse light: polarisation lost → partly passes

- A **polariser** in front of the light, and a second one — the **analyser** — in front of the lens, turned at 90°.
- A mirror reflection keeps its polarisation, so the analyser **blocks it**. Light scattered inside a matt or coloured surface loses its polarisation and partly passes.
- The glare disappears, the surface underneath appears. The price: the image is darker — each polariser removes at least half of the light.
- Light reflected by glass, plastic or water is already partly polarised: an analyser alone on the lens removes much of it.

USED FOR

Labels under shrink film and blister packs · food with shiny skins · wet surfaces · codes on plastic.

Ultraviolet light and fluorescence

Light at one wavelength in, a glow at another out

- **Fluorescence:** some materials absorb UV (typically 365 nm, "black light") and re-emit visible light. A filter on the lens blocks the UV, so the camera sees **only what glows**.
- It is "non-linear" in the sense that the light seen has a different colour from the light sent: reflections of the lamp are filtered out completely.
- **Finer detail:** the smallest detail a lens can resolve is about $\lambda / (2\,\mathrm{NA})$. Shorter wavelengths resolve finer detail — one reason UV is used in microscopy and wafer inspection.

What glows	Inspection
fluorescent penetrant drawn into cracks	crack detection in castings, welds, forgings (ISO 3452)
invisible UV inks	marks, codes, security features
adhesives, varnishes, coatings with a tracer	is the glue there, and where?
oils, grease, organic residues	cleanliness of parts
fluorescent tracer particles	flow measurement (PIV): only the particles shine, not the walls

UV-A sources and the eyes: UV lamps need shields and UV-blocking glasses for the operators.

Infrared, structured light, strobes

Three more ways to control what the camera sees

Near infrared (850 nm)

Invisible to operators, so they are not dazzled. With a band-pass filter on the lens, the room's light and the sun are almost removed: a lighting that stays the same all day. Many colour inks are transparent in the infrared: it shows what is under a print, and turns coloured parts into plain grey.

Structured light

Light with a known pattern in space — a laser line, a grid of dots, projected fringes. The way the pattern bends on the part gives its **3D shape**: the subject of the 3D shape measurement lecture.

Strobe (flash)

Light in time: a pulse of microseconds, synchronised with the camera, freezes a moving part — 82 µs on a conveyor at 0.5 m/s (lecture 7). Overdriven LEDs give pulses many times brighter than their continuous rating.

Keep the light constant

Every threshold assumes the light of the day it was set. Warm the light up, shield it from the room, monitor its level — a photodiode near the source is enough — and check a reference image at every shift start. A light that drifts is a gauge that drifts.

Choosing the light

Start from what must be seen

You need to see…	Light	Why
outline, diameters, holes	**backlight** — collimated for precision	silhouette, independent of colour and finish
scratches, dust, cracks, embossing	**dark field** / low-angle ring	only the defect scatters light into the lens
engraving or print on flat shiny metal	**coaxial**	flat is bright, tilted is dark
curved shiny parts	**dome** / cloudy-day	no reflection of the lamp
matt parts in general	**ring** or diffuse area light	even, few shadows
a colour against another	coloured light + monochrome camera	one colour dark, the other bright
a label through glare or film	**polarisers**	mirror reflections blocked
cracks, glue, residues	**UV** + fluorescence	only the target glows
a moving part; a changing room	**strobe**; near-IR + filter	freeze motion; ignore ambient light

Recap TAKEAWAYS

Parts 3 and 4 — geometry and special techniques

1. **Bright field or dark field:** does the mirror reflection go into the camera? Dark field makes scratches and dust shine on black.
2. **Ring, coaxial, dome:** general purpose; flat shiny with engravings; curved shiny without reflections.
3. **Backlight** gives the silhouette for dimensions; collimated with a telecentric lens for precision.
4. **Polarisers** block glare; **UV** makes cracks, glue and residues glow; **near-IR** ignores the room and sees through prints; **strobes** freeze motion.
5. **One light per feature**, chosen from what must be seen — and kept constant, because a drifting light is a drifting gauge.

Next
Interest points: finding the same feature in two images, whatever its size, rotation and light.

The hands-on lab
You build a backlight from a phone screen and a sheet of paper.

Questions?

Prof. Gianluca Rossi • dr. Lorenzo Capponi

lorenzo.capponi@collaboratori.unipg.it • lorenzocapponi@outlook.it

github.com/LolloCappo/student-hub